PRACTICAL SET – 1

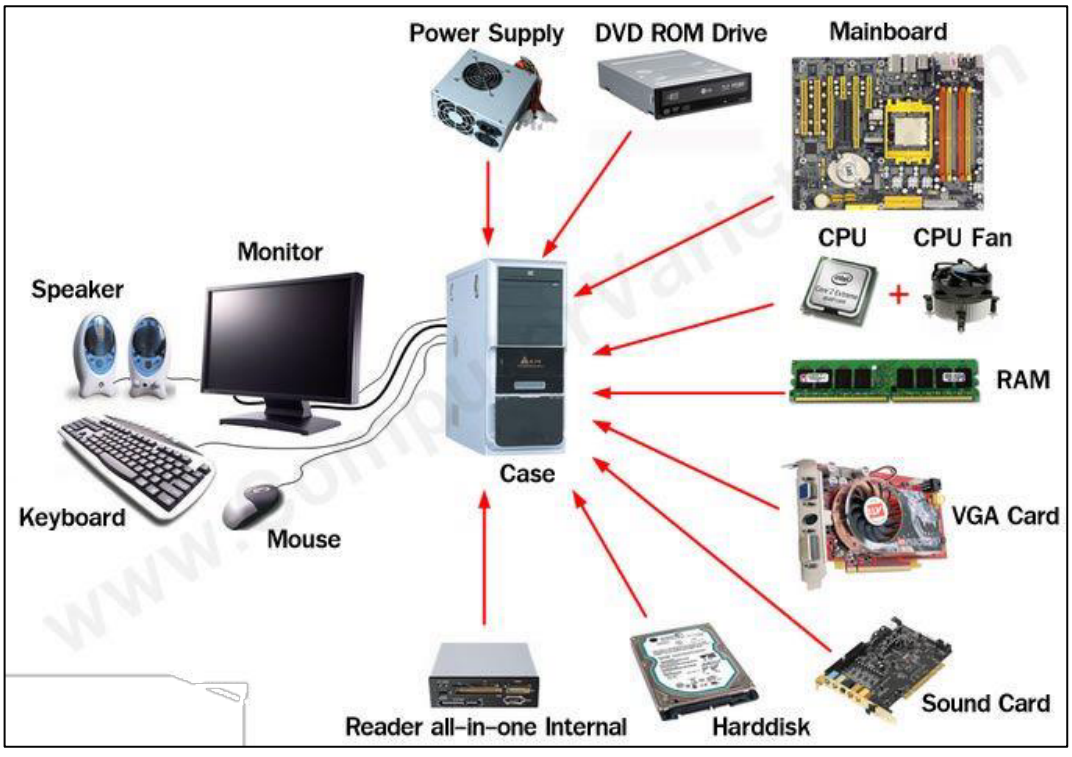
1. Introduction of computer systems and operating systems (windows, Linux, Mac).

A. **Components of computer system**
B. **Compiler and interpreter**
C. **Concepts of Machine level, Assembly level and high-level programming Language**
D. **Concepts of Hardware, hardware peripherals like RAM, ROM, keyboard, Mouse, processors, etc; Working of SMPS**
E. **Study of various ports, Steps and precautions to assemble computer**

A. Components of computer system:

- A computer system consists of both hardware and software components that work together to perform tasks. These components can be broadly classified into:

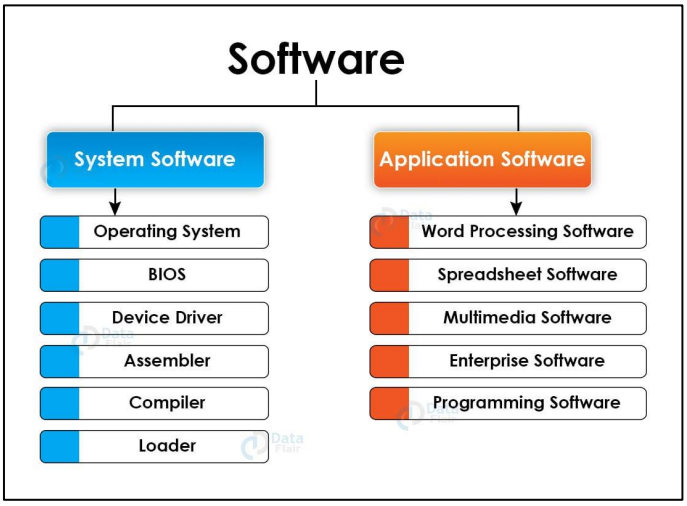
1. **Hardware**: The physical components of the computer, including:



- **Central Processing Unit (CPU)**: The brain of the computer, responsible for executing instructions and performing calculations.

- **Memory (RAM & ROM)**: Temporary and permanent storage of data, with RAM being volatile (data is lost when the computer is turned off) and ROM being non-volatile (data is preserved even when the computer is powered off).
- **Storage Devices**: Hard Drive (HDD), Solid-State Drive (SSD), Optical Drives (CD/DVD), etc., used for long-term data storage.
- **Input Devices**: Devices like keyboard, mouse, microphone, etc., used to input data into the system.
- **Output Devices**: Devices like monitors, printers, and speakers that display or provide output from the system.
- **Motherboard**: The main circuit board that holds the CPU, memory, and other essential components, and allows communication between them.
- **Power Supply Unit (PSU)**: Supplies electrical power to the computer's internal components.

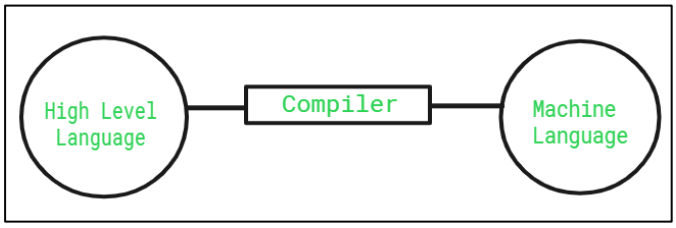
2. **Software**: The programs and data that run on the computer, including:



- **Operating System (OS)**: Manages hardware resources and provides an interface for user interaction. Examples include Windows, Linux, and macOS.
- **Applications**: Programs designed to perform specific tasks, like web browsers, word processors, games, etc.

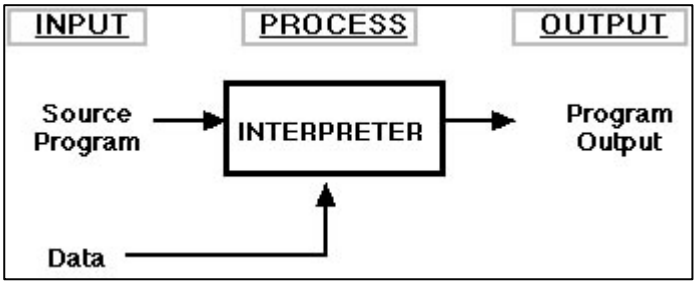
B. Compiler and Interpreter

❖ Both compilers and interpreters are tools used to translate high-level programming code into machine code, but they work in different ways:



1. **Compiler**:
    - A compiler translates the entire source code of a program into machine code (binary code) all at once before execution.
    - The output is a standalone machine code file (e.g., an executable file like .exe in Windows).
    - Example languages: C, C++, Rust.
    - **Pros**: Faster execution, as the translation is done beforehand.
    - **Cons**: Debugging can be more difficult, and the compilation process takes longer
2. **Interpreter**:

An interpreter translates high-level code into machine code line by line, executing each instruction immediately.



- The source code remains in its original form and is interpreted every time it is run.
- Example languages: Python, Ruby, JavaScript.
- **Pros**: Easier to debug and more interactive.
- **Cons**: Slower execution because translation happens at runtime.

C. Concepts of Machine Level, Assembly Level, and High-Level Programming Languages

1. **Machine-Level Language**:

    - The lowest level of programming language, consisting of binary code (0s and 1s).
    - Directly understood by the computer's CPU.
    - Not human-readable, and programming in this language is complex and error-prone.

2. **Assembly-Level Language**:
    - A step above machine language, using mnemonics (symbols) instead of binary code (e.g., MOV for moving data).
    - Requires an assembler to convert assembly code into machine code.

- Easier to read and write than machine-level language but still closely tied to hardware architecture.

3. **High-Level Programming Language**:
    - A language closer to human languages, abstracted from the hardware.
    - Easier to write, read, and maintain than assembly or machine code.
    - Examples include C, Python, Java, JavaScript, and Ruby.
    - High-level languages are platform-independent, meaning the same program can run on different operating systems or hardware with minimal changes.

D. Concepts of Hardware, Hardware Peripherals, and Working of SMPS

1. **Hardware**: Physical components that make up a computer system, as discussed earlier.
2. **Hardware Peripherals**:
    - **RAM (Random Access Memory)**: Temporary storage used by the CPU to store data that is actively being processed. It is volatile and loses data when the system is powered off.
    - **ROM (Read-Only Memory)**: Non-volatile storage used to store firmware, such as the BIOS, which is critical for booting the computer.
    - **Keyboard and Mouse**: Input devices that allow the user to interact with the computer.
    - **Monitor**: An output device that displays visual information generated by the computer.
    - **Processor (CPU)**: The heart of the computer, performing calculations and executing instructions. It consists of the Control Unit (CU) and Arithmetic Logic Unit (ALU).
3. **SMPS (Switched-Mode Power Supply)**:
    - SMPS is responsible for converting the electrical power from a wall outlet (AC power) into a form usable by the computer's components (DC power).
    - It is more efficient than traditional linear power supplies, generating less heat and consuming less energy.
    - The SMPS ensures that the computer receives the correct voltage and current for each component, such as 12V for the CPU and 5V for peripherals.

E. Study of Various Ports, Steps, and Precautions to Assemble a Computer

1. **Types of Ports**:
    - **USB Ports**: Used for connecting a wide range of devices, including keyboards, mice, printers, external drives, etc.
    - **HDMI/DisplayPort**: For video and audio output to monitors or projectors.
    - **Audio Jacks**: For connecting headphones, speakers, or microphones.
    - **Ethernet (LAN) Port**: Used for wired network connections.
    - **VGA/DVI Ports**: Older video output standards (rarely used now but may be found on legacy systems).

- **PCI Slots**: Expansion slots for adding hardware components such as graphics cards or sound cards.

2. **Steps to Assemble a Computer**:
   - **Preparation**: Gather all the components (motherboard, CPU, RAM, storage devices, power supply, etc.).
   - **Install the CPU**: Place the CPU into the motherboard socket, making sure it is aligned correctly.
   - **Install RAM**: Insert RAM sticks into the motherboard slots, ensuring they are securely clicked into place.
   - **Install the Motherboard**: Mount the motherboard inside the computer case and screw it into place.
   - **Connect Power Supply**: Attach the power supply to the motherboard and other components.
   - **Install Storage Devices**: Connect the hard drive or SSD to the motherboard using the appropriate cables (SATA or NVMe).
   - **Install GPU (if applicable)**: Insert the graphics card into the PCIe slot on the motherboard.
   - **Connect Peripherals**: Attach the keyboard, mouse, monitor, and other input/output devices.
   - **Check Connections**: Ensure that all components are properly connected and seated.
   - **Power On and Test**: Turn on the system and ensure that it boots successfully.

3. **Precautions While Assembling a Computer**:
   - **Static Electricity**: Use an anti-static wristband or frequently ground yourself to avoid damaging sensitive components with electrostatic discharge (ESD).
   - **Handle Components Carefully**: Avoid touching the pins and connectors on components, especially the CPU and motherboard.
   - **Proper Airflow**: Ensure that the components, especially the CPU and GPU, are adequately cooled to prevent overheating.
   - **Cable Management**: Tidy up cables to prevent obstruction of airflow and ensure a neat, efficient system.

PRACTICAL SET – 2

2. Demonstrate the various commands of Linux (ls, pwd, mkdir, cd, rmdir, cp, mv, rm, cat, ps, man, grep, echo, sort, cal etc).

pwd Command

The pwd command is used to display the location of the current working directory.

Syntax:

1. pwd

Output:

```
javatpoint@javatpoint-Inspiron-3542:~$ pwd
/home/javatpoint
```

mkdir Command

The mkdir command is used to create a new directory under any directory.

Syntax:

1. mkdir **<directory** name>

Output:

```
javatpoint@javatpoint-Inspiron-3542:~$ mkdir new_directory
javatpoint@javatpoint-Inspiron-3542:~$
```

rmdir Command

The rmdir command is used to delete a directory.

Syntax:

1. rmdir **<directory** name>

Output:

```
javatpoint@javatpoint-Inspiron-3542:~$ rmdir new_directory
javatpoint@javatpoint-Inspiron-3542:~$
```

ls Command

The ls command is used to display a list of content of a directory.

Syntax:

1. ls

Output:

```
javatpoint@javatpoint-Inspiron-3542:~$ ls
a               Desktop             examples.desktop  Music          sample
Akash           Directory           hello.c           pico           snap
a.out           Documents           hello.i           Pictures       Templates
composer.phar   Downloads           hello.o           project        Test.txt
Demo.sh         eclipse             hello.s           Public         Videos
Demo.txt        eclipse-installer   index.html        Python
Demo.txt~       eclipse-workspace   mail              Python-3.8.0
```

cd Command

The cd command is used to change the current directory.

Syntax:

1. cd <directory name>

Output:

```
javatpoint@javatpoint-Inspiron-3542:~$ cd Desktop
javatpoint@javatpoint-Inspiron-3542:~/Desktop$
```

cat Command

The cat command is a multi-purpose utility in the Linux system. It can be used to create a file, display content of the file, copy the content of one file to another file, and more.

Syntax:

1. cat [OPTION]... [FILE]..

To create a file, execute it as follows:

1. cat > <file name>
2. // Enter file content

   Press "**CTRL+ D**" keys to save the file. To display the content of the file, execute it as follows:

1. cat <file name>

Output:

```
javatpoint@javatpoint-Inspiron-3542:~/Newfolder$ cat > Demo.txt
This is a text file.
javatpoint@javatpoint-Inspiron-3542:~/Newfolder$ cat Demo.txt
This is a text file.
```

rm Command

The rm command is used to remove a file.

Syntax:

rm <file name>

Output:

```
javatpoint@javatpoint-Inspiron-3542:~/Newfolder$ rm Demo.txt
javatpoint@javatpoint-Inspiron-3542:~/Newfolder$ rm Demo1.txt Demo2.txt
```

cp Command

The cp command is used to copy a file or directory.

Syntax:

To copy in the same directory:

1. cp <existing file name> <new file name> To copy in a different directory:

Output:

```
javatpoint@javatpoint-Inspiron-3542:~$ cp demo.txt demo1.txt
javatpoint@javatpoint-Inspiron-3542:~$ cp demo.txt Documents
```

mv Command

The mv command is used to move a file or a directory form one location to another location

Syntax:

1. mv <file name> <directory path>

Output:

```
javatpoint@javatpoint-Inspiron-3542:~$ mv demo.txt Directory
```

grep Command

The grep is the most powerful and used filter in a Linux system. The 'grep' stands for "**global regular expression print**." It is useful for searching the content from a file. Generally, it is used with the pipe.

Syntax:

1. command | grep **<searchWord>**

Output:

```
javatpoint@javatpoint-Inspiron-3542:~$ cat marks.txt | grep 9
celena-90
```

cal Command

The cal command is used to display the current month's calendar with the current date highlighted.

Syntax:

1. cal<

Output:

```
javatpoint@javatpoint-Inspiron-3542:~$ cal
      May 2020
Su Mo Tu We Th Fr Sa
                1  2
 3  4  5  6  7  8  9
10 11 12 13 14 15 16
17 18 19 20 21 22 23
24 25 26 27 28 29 30
31
```

ps command in Linux

ps command in Linux is used to check the active processes in the terminal.

For each process, a unique identification number, known as the PID, is displayed. The TTY column shows the terminal device that is associated with each process. The TIME column displays the amount of CPU time used by each process.

The CMD column shows the executable command that was used to start each process.

Command:

```
1  ps
```

Output:

```
PID TTY          TIME CMD
8454 pts/521  00:00:00 bash
11982 pts/521  00:00:00 bash
11983 pts/521  00:00:00 ps
```

man command in Linux

The **man command** displays a user manual for any commands or utilities available in the Terminal, including their name, description, and options.

Command to view the full manual:

man <command name>

For example, suppose you want to look up the manual for the ls command: **man ls**

Command:

```
1  man -f ls
```

Output:

```
ls (1) - list directory contents
```

sort command in Linux

The **sort** command is used generally to sort the output of the file. Let's use the command and see the output.

Command: (We are using the cat command to

```
1  cat multiple.txt
```

Output: (The content of multiple.txt file in the terminal)

```
Hello World
GeeksforGeeks
Thank you
```

Now we will sort the outcome using the **sort command**

Command:

```
1  sort multiple.txt
```

Output:

```
GeeksforGeeks
Hello World
Thank you
```

- Here first we checked the file content using the **cat** command and then we sorted it alphabetically using the **sort** command.

echo command in Linux

echo command in Linux is specially used to print something in the terminal

Command:

```
1  echo "Hello World"
```

Output:

```
Hello World
```

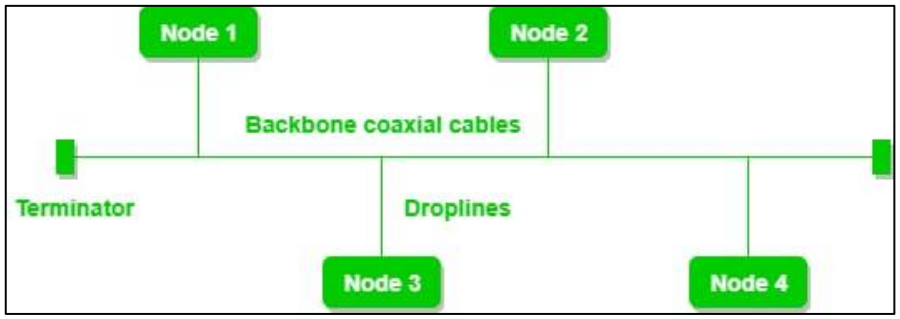
PRACTICAL SET – 3

3. **Identify Various Computer Network Tools:**
   a. **Various Network Topologies**
   b. **Different types of network cables and practically implement the cross wired cable and straight through cable using a crimping tool.**
   c. **Basic Network Commands**

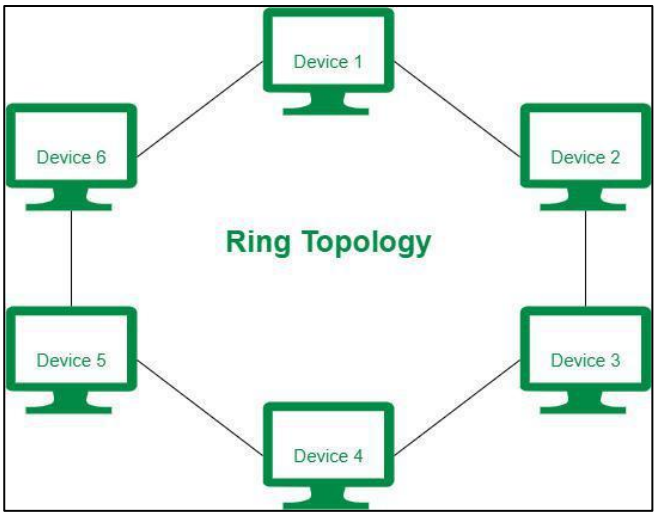
A. Various Network Topologies

- Network topologies define the structure or arrangement of elements (links, nodes, etc.) in a computer network. Common network topologies include:

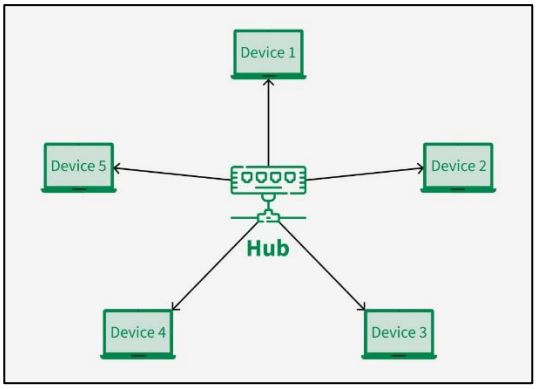
1. **Bus Topology**:
   - All devices are connected to a central cable (the bus).
   - Simple and cost-effective but limited by cable length and number of devices.



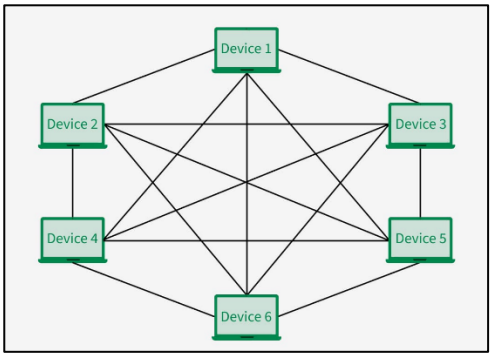
2. **Ring Topology**:
   - Devices are connected in a circular or ring shape.
   - Data travels in one direction; each device acts as a repeater to boost the signal.



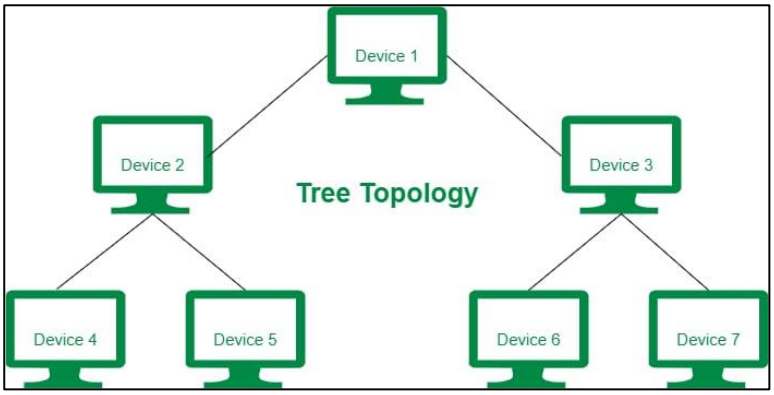
3. **Star Topology**:
    - All devices are connected to a central device (like a switch or hub).
    - Easy to manage and scalable, but failure of the central device can bring down the network.



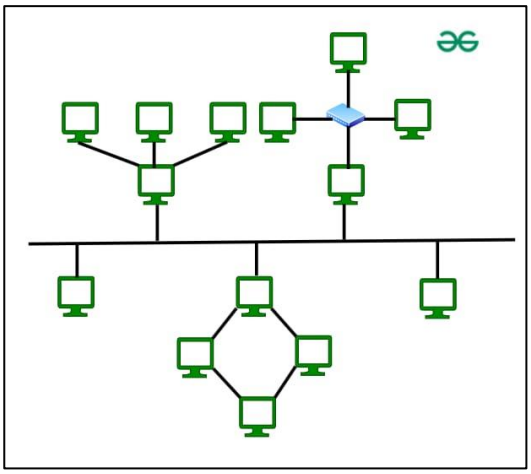
4. **Mesh Topology**:
    - Every device is connected to every other device, providing redundancy and fault tolerance.
    - Expensive and complex but very reliable.



5. **Tree Topology**:
    - A hybrid topology that combines characteristics of star and bus topologies.
    - Devices are grouped into star-configured groups, and these groups are connected by a central bus backbone.



6. **Hybrid Topology**:
    - A combination of two or more different topologies to leverage the benefits of each.
    - Can be designed to meet specific requirements of large networks.



B. Different Types of Network Cables and Practically Implementing Cross-Wired and Straight-Through Cables Using a Crimping Tool

➢ **Types of Network Cables**:

1. **Twisted Pair Cables**:
    - **Unshielded Twisted Pair (UTP)**: Most commonly used in Ethernet networks.
    - **Shielded Twisted Pair (STP)**: Used where there's higher interference; it has an extra layer of shielding.
2. **Coaxial Cable**:
    - Older type of cable used in earlier Ethernet networks.
    - Has a central conductor, an insulating layer, a shield, and an outer layer.
3. **Fiber Optic Cable**:
    - Uses light to transmit data, making it ideal for long-distance and high-speed networking.
    - Types: Single-mode and multi-mode.

Crimping Tool Use:

- A crimping tool is used to attach connectors (RJ45 or RJ11) to the ends of twisted-pair cables.

➢ **Types of Ethernet Cables**:

1. **Straight-Through Cable**:
    - Used to connect different types of devices, such as a computer to a switch or router.

- The wiring is the same on both ends (e.g., both ends use the same color sequence like T568A or T568B).

Wiring for Straight-Through:

- Pin 1: White/Blue
- Pin 2: Blue
- Pin 3: White/Orange
- Pin 4: Orange
- Pin 5: White/Green
- Pin 6: Green
- Pin 7: White/Brown
- Pin 8: Brown

2. **Crossover Cable**:
   - Used to connect devices of the same type, such as a computer to another computer or a switch to a switch.
   - In a crossover cable, the wiring is different on each end (one end uses T568A and the other uses T568B).

Wiring for Crossover:

- One end is wired according to T568A and the other end according to T568B.

Steps to Crimp a Cable:

- **Cut the Cable**: Use a wire cutter to get the required length of the cable.
- **Strip the Insulation**: Use a wire stripper to remove about 1.5 inches of the outer insulation.
- **Arrange the Wires**: Arrange the individual wires in the correct order (straight-through or crossover).
- **Insert into RJ45 Connector**: Insert the arranged wires into an RJ45 connector.
- **Crimp**: Use a crimping tool to press the connector onto the cable, ensuring the pins penetrate the wires.

C. Basic Network Commands

1. **ping**:
   - Used to test the connectivity to another network device.
   - Example: `ping 192.168.1.1`
2. **ipconfig (Windows) / ifconfig (Linux/macOS)**:
   - Displays the current network configuration (IP address, subnet mask, default gateway).
   - Example: `ipconfig` (Windows) or `ifconfig` (Linux/macOS)
3. **tracert (Windows) / traceroute (Linux/macOS)**:

- Traces the path data takes to reach a destination, showing all intermediate devices (routers) along the way.
- Example: `tracert google.com`

4. **nslookup**:
   - Used to query DNS servers to get domain name or IP address information.
   - Example: `nslookup www.example.com`
5. **netstat**:
   - Shows network statistics, including active connections and listening ports.
   - Example: `netstat -a`
6. **route**:
   - Displays or modifies the network routing table.
   - Example: `route print`
7. **ip route** (Linux):
   - Shows or manipulates the routing table in Linux-based systems.
   - Example: `ip route show`
8. **netsh** (Windows):
   - A command-line tool used to configure network interfaces, firewall settings, and more.
   - Example: `netsh interface ip show config`
9. **arp**:
   - Used to display or modify the ARP (Address Resolution Protocol) cache.
   - Example: `arp -a`
10. **ftp**:
    - A command-line tool to transfer files between hosts using the FTP protocol.
    - Example: `ftp ftp.example.com`

PRACTICAL SET – 4

4. Basic concept of various electronic devices, Internet Things (IoT), its working architectures, sensors and IoT boards.

❖ The **Internet of Things (IoT)** is one of the most transformative concepts in modern **Information and Communication Technology (ICT)**. It refers to the interconnection of everyday physical devices—ranging from household appliances to industrial machines—via the **internet** to collect and exchange data. The key to IoT's success lies in its ability to gather data from the environment, process it, and act on it, without human intervention. To understand how this works, it is important to know about the various **electronic devices**, **sensors**, **IoT boards**, and **architectures** involved in IoT systems.

1. Basic Concept of Electronic Devices in IoT

Electronic devices are the physical components that enable interaction with the environment and the digital world. In the context of **IoT**, these devices are typically embedded with sensors and actuators, connected to networks, and are capable of performing data collection and communication tasks.

- **Sensors**: Devices that detect changes in the environment (such as temperature, humidity, motion, etc.) and convert them into electrical signals.
- **Actuators**: Devices that respond to data by acting upon the environment (e.g., turning on a light, opening a valve, etc.).
- **Microcontrollers**: The "brains" of an IoT device, microcontrollers process the sensor data and control the actuators based on predefined algorithms.

2. What is the Internet of Things (IoT)?

The **Internet of Things (IoT)** refers to the network of physical objects (devices, sensors, actuators, and other items) that are embedded with sensors, software, and other technologies to collect and exchange data over the internet. The main goal of IoT is to connect the physical world to the digital world, enabling automation, real-time monitoring, and remote control of devices.

Examples of IoT Devices:

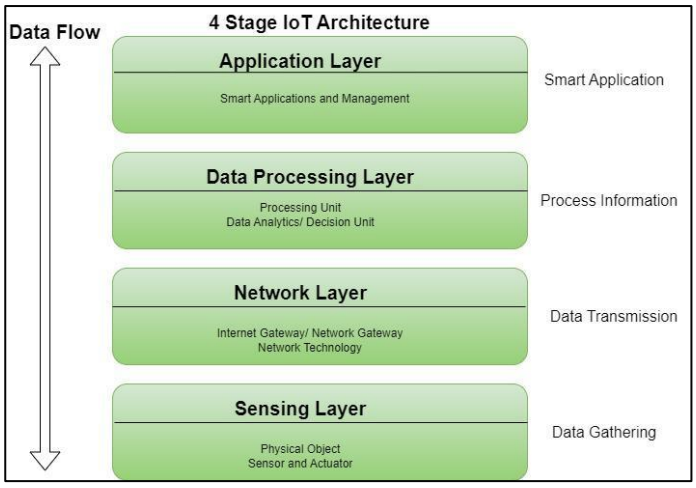
- **Smart Homes**: Smart thermostats (e.g., Nest), smart lights (e.g., Philips Hue), security cameras.
- **Healthcare**: Wearable fitness trackers (e.g., Fitbit), remote patient monitoring devices.
- **Industrial IoT (IIoT)**: Sensors in machinery for predictive maintenance, smart factories.

3. Working Architectures of IoT

The architecture of an IoT system defines how data flows from the physical world (sensors and actuators) to digital processing units (cloud, servers) and back. There are several types of IoT architectures based on the application and requirements.

- **Common IoT Architecture Layers:**

1. **Perception Layer**:
    - The perception layer, also known as the **sensing layer**, is the foundation of an IoT system. It consists of physical devices (sensors, actuators) that collect data from the environment.
    - **Devices**: Temperature sensors, motion sensors, GPS modules, cameras, humidity sensors, etc.
2. **Network Layer**:
    - The network layer transmits the data collected from the perception layer to the processing layer (cloud or local servers) through various communication protocols.
    - **Protocols**: Wi-Fi, Bluetooth, Zigbee, LoRaWAN, 5G, and others are commonly used to send data from IoT devices to the cloud.
3. **Processing Layer**:
    - The processing layer processes the data received from the network layer. This is where data is analyzed, stored, and decisions are made. Typically, the data is sent to cloud servers where advanced computing, AI algorithms, and data analytics are used to make sense of it.
    - **Cloud Platforms**: AWS IoT, Google Cloud IoT, Microsoft Azure IoT are some examples of cloud platforms used to manage and process IoT data.
4. **Application Layer**:
    - The application layer is where end-users interact with the system. This is typically the layer where the processed data is displayed or acted upon in real-time. Applications can be for industries such as agriculture, healthcare, transportation, etc.
    - **Example**: In a smart home application, the user interacts with a mobile app to control lights, heating, or security cameras.
5. **Business Layer**:
    - The business layer focuses on business logic and managing operations of the IoT system. It can include decision-making algorithms, billing systems, or resource management.



4. IoT Sensors and Their Types

Sensors are one of the most essential components of IoT devices. They capture data from the physical world and convert it into a digital format that can be processed.

Types of IoT Sensors:

1. **Temperature Sensors**:
   - Detect the temperature of an object or environment (e.g., **DHT11**, **LM35**).
   - Applications: Home automation (smart thermostats), industrial temperature monitoring.
2. **Humidity Sensors**:
   - Measure the humidity levels in the air (e.g., **DHT22**, **HIH-4000**).
   - Applications: Weather monitoring, smart agriculture, HVAC systems.
3. **Motion Sensors**:
   - Detect motion or presence (e.g., **PIR sensors**).
   - Applications: Security systems, automated lighting systems.
4. **Pressure Sensors**:
   - Measure the pressure of gases or liquids (e.g., **BMP180**).
   - Applications: Weather stations, industrial IoT for pressure monitoring.
5. **Proximity Sensors**:
   - Detect the presence of objects or people (e.g., **Ultrasonic sensors**).
   - Applications: Robotics, smart parking, manufacturing automation.
6. **Light Sensors**:
   - Measure light intensity (e.g., **LDR** or Light Dependent Resistor).
   - Applications: Smart lighting systems, outdoor environment monitoring.
7. **Gas Sensors**:
   - Detect the presence of gases (e.g., **MQ series** for gases like CO, methane, alcohol).
   - Applications: Air quality monitoring, industrial leak detection.

8. **Sound Sensors**:
    - Detect sound levels (e.g., **microphone sensors**).
    - Applications: Noise monitoring, security systems.

5. IoT Boards and Microcontrollers

IoT boards (or development boards) are specialized electronic boards designed for IoT applications. These boards often come with built-in Wi-Fi, Bluetooth, and other networking capabilities to help developers build IoT systems more easily.

➢ **Common IoT Boards and Microcontrollers:**

1. **Arduino**:
    - **Arduino Uno** is one of the most popular development boards for building IoT projects. It is based on the **ATmega328** microcontroller and can be used with various sensors and actuators.
    - Arduino has a large community and extensive libraries, making it easy to get started with IoT projects.
2. **Raspberry Pi**:
    - Raspberry Pi is a small, affordable computer that runs a full operating system (usually **Raspberry Pi OS**) and supports programming languages like Python, JavaScript, etc.
    - Raspberry Pi can act as a powerful hub for IoT systems, handling complex processing tasks and running multiple applications simultaneously.
3. **ESP8266/ESP32**:
    - **ESP8266** is a low-cost Wi-Fi module, widely used in IoT applications. It is often paired with **Arduino** to create simple IoT solutions.
    - **ESP32** is a more powerful successor to the ESP8266 with built-in **Wi-Fi** and **Bluetooth** capabilities, often used for more complex applications.
4. **Intel Edison**:
    - A small, powerful development board designed for IoT applications. It runs a version of **Linux** and supports multiple programming languages.
    - Often used for industrial IoT solutions due to its high processing power.
5. **BeagleBone**:
    - BeagleBone is a single-board computer similar to Raspberry Pi, but it is often preferred for more industrial and embedded IoT applications.
    - Supports Linux-based operating systems and is often used for real-time applications.

6. IoT Communication Protocols

Communication is a vital component of any IoT system. Devices need to communicate over the internet, which is done using various communication protocols.

- **Popular IoT Communication Protocols:**

  - **MQTT (Message Queuing Telemetry Transport)**: A lightweight messaging protocol ideal for low-bandwidth, high-latency networks.
  - **HTTP/HTTPS**: Standard web protocols, often used for communication between IoT devices and web servers.
  - **CoAP (Constrained Application Protocol)**: A specialized protocol designed for constrained devices and networks.
  - **LoRaWAN (Long Range Wide Area Network)**: A low-power, wide-area network protocol designed for long-range communication between IoT devices.
  - **Zigbee**: A short-range communication protocol used for low-power IoT devices in home automation.

PRACTICAL SET – 5

5. Study of evaluation from 2G to 5G and its use cases and various internet-based services like email, social network, chat, etc.

- The evolution from **2G to 5G** represents a monumental leap in the field of Information and Communication Technology (ICT). Each new generation of mobile networks has significantly improved speed, connectivity, and functionality, paving the way for more advanced internet-based services like **email**, **social networks**, **chat applications**, and other **cloud-based services**. The role of mobile networks has expanded from simple voice communication to enabling a fully interconnected digital ecosystem that supports IoT (Internet of Things), high-definition video, real-time communication, and even augmented/virtual reality.

➢ 2G (Second Generation) – The Digital Revolution in Mobile Communication

Launched: Early 1990s
Key Technologies: GSM (Global System for Mobile Communications), CDMA (Code Division Multiple Access)

➢ **Key Features and Use Cases:**

- **Voice Communication**: 2G technology was primarily designed for voice services. It provided better quality than the analog networks of the previous generation (1G).
- **SMS (Short Message Service)**: Introduction of **SMS** enabled users to send short text messages, which rapidly became a popular communication tool.
- **Limited Data Transmission**: 2G offered basic data services (up to **14.4 kbps**), which were too slow for multimedia content but sufficient for simple text-based services like email and browsing basic websites.
- **Roaming**: Enabled international roaming, allowing users to use their phones in different countries.

➢ **Internet-Based Services:**

- **Email**: Limited mobile access to email, usually text-based and without multimedia attachments. Email services were primarily designed for desktop usage.
- **Social Networks**: Social networks were not prevalent during the 2G era. The internet was in its infancy, and social interaction was mostly via email and early online forums.
- **Chat**: The only form of mobile chat was SMS, which was limited to simple text messages.
- **Web Browsing**: Extremely slow and basic browsing. The mobile internet was used primarily for informational purposes, and websites were text-heavy with limited graphics.

➢ **Limitations:**

- Data speeds were insufficient for advanced multimedia services.
- The internet experience was very basic, restricting user interaction with online content.

➢ 3G (Third Generation) – The Era of Mobile Broadband

Launched: Early 2000s
Key Technologies: WCDMA (Wideband Code Division Multiple Access), HSPA (High-Speed Packet Access)

➢ Key Features and Use Cases:

- **Higher Data Speeds**: 3G networks significantly increased data speeds to **384 kbps – 2 Mbps**, making mobile internet access much more practical.
- **Video Calls**: With 3G, **video calling** became possible, a significant innovation that allowed users to interact face-to-face remotely.
- **Improved Web Browsing**: 3G provided much faster browsing capabilities, allowing for richer web experiences with images, text, and multimedia content.
- **Mobile Hotspot**: Enabled devices to share their mobile internet connection, making it possible for other devices (like laptops) to access the internet via the mobile network.

➢ Internet-Based Services:

- **Email**: Mobile email became more accessible, with services like **Gmail** and **Yahoo Mail** providing push notifications and the ability to download attachments.
- **Social Networks**: Early versions of **Facebook**, **Twitter**, and **LinkedIn** became available on mobile, but with limited functionality. Users could view posts, comment, and upload basic media.
- **Chat**: Messaging apps like **MSN Messenger**, **Yahoo Messenger**, and SMS-based chat services began to grow in popularity.
- **Web Browsing**: 3G enabled mobile users to browse websites with richer content, including multimedia files, although the experience was still slow compared to desktop browsing.

➢ Limitations:

- While much faster than 2G, 3G speeds were still insufficient for high-definition streaming or large file transfers.
- Network congestion in busy areas still impacted the quality of service.

➢ 4G (Fourth Generation) – The Age of High-Speed Internet and Mobile Multimedia

Launched: Late 2000s to early 2010s
Key Technologies: LTE (Long Term Evolution), WiMAX

- **Key Features and Use Cases:**

  - **Blazing Fast Data Speeds**: 4G networks offered speeds of **100 Mbps to 1 Gbps**, making it possible to stream **HD videos**, download large files, and use data-intensive applications on mobile devices.
  - **VoLTE (Voice over LTE)**: Enabled high-quality voice calls over the 4G network, replacing older circuit-switched methods.
  - **Mobile Gaming**: The rise of cloud-based gaming services and mobile gaming on smartphones was made possible by the increased bandwidth and low latency of 4G.
  - **Mobile Video Streaming**: High-definition video streaming from platforms like **Netflix**, **YouTube**, and **Amazon Prime** became mainstream.

- **Internet-Based Services:**

  - **Email**: With high-speed internet, email became fully integrated with cloud-based services like **Google Workspace** and **Microsoft 365**, allowing users to access emails, documents, and calendars in real-time across multiple devices.
  - **Social Networks**: Social media apps like **Facebook**, **Instagram**, **Twitter**, and **Snapchat** evolved to support video sharing, live streaming, and other interactive content.
  - **Chat**: Instant messaging apps such as **WhatsApp**, **Telegram**, **Facebook Messenger**, and **iMessage** gained widespread popularity, with support for text, voice messages, video calls, and multimedia sharing.
  - **Web Browsing**: Mobile browsing became as fast and rich as on desktop computers, with HTML5, interactive websites, and robust media capabilities.
  - **Video Conferencing**: Services like **Zoom**, **Skype**, and **Google Meet** gained traction for remote work and online education due to the improved data speeds and quality of service.

- **Limitations:**

  - 4G networks were still relatively limited in rural areas, and network congestion could occasionally affect speeds during peak usage times.
  - Despite significant improvements, network coverage was not universal, and rural areas often experienced slower speeds.

➢ 5G (Fifth Generation) – The Next Frontier in Connectivity and Innovation

Launched: 2019 (Commercial rollouts began globally)
Key Technologies: mmWave, Sub-6 GHz, Massive MIMO, Beamforming, Network Slicing

- **Key Features and Use Cases:**

  - **Ultra-High Speeds**: 5G promises speeds up to **20 Gbps**, making it possible to download **4K/8K video** in seconds and enabling **real-time collaboration** with no delays.
  - **Ultra-Low Latency**: Latency as low as **1 millisecond** enables **autonomous vehicles**, **remote surgeries**, and **augmented reality (AR)/ virtual reality (VR)** applications that require instant response times.
  - **Massive Connectivity**: Supports up to **1 million devices per square kilometer**, ideal for **IoT** (Internet of Things) applications, including smart homes, smart cities, and industrial automation.
  - **Network Slicing**: Enables the creation of virtualized networks for different use cases (e.g., low-latency for autonomous vehicles, high throughput for AR/VR).

- **Internet-Based Services:**

  - **Email**: With **5G**, email synchronization is near-instantaneous, and large attachments can be downloaded or uploaded without delays. Real-time collaboration in cloud services becomes seamless.
  - **Social Networks**: Social media platforms will evolve with features like **real-time 4K video streaming**, **advanced AR filters**, and **instant content sharing**. **TikTok**, **Snapchat**, and **Instagram** could integrate enhanced AR/VR experiences.
  - **Chat**: Enhanced messaging apps will support **ultra-high-definition video calls**, instantaneous file sharing (including large videos), and real-time collaboration, even for remote teams working on large media projects.
  - **Web Browsing**: 5G will allow **instant browsing** with richer, interactive content. Websites will support immersive experiences like **AR** and **VR**, providing users with a fully interactive, multimedia-rich internet experience.
  - **Cloud Gaming**: The latency improvements of 5G will make mobile cloud gaming services like **Google Stadia**, **Xbox Cloud Gaming**, and **NVIDIA GeForce NOW** seamless, enabling high-quality games to be played on any device.
  - **Video Conferencing**: High-definition video conferencing will be the norm, with **5G** enabling **4K/8K video calls** with no buffering or latency, revolutionizing remote work, education, and healthcare services.

- **Limitations:**

  - **5G Rollout**: While 5G promises substantial improvements, its rollout is still ongoing, and global coverage remains limited, especially in rural areas.
  - **Infrastructure**: 5G requires the deployment of new infrastructure like **small cells**, and its millimeter-wave spectrum requires a denser network of cell towers.

- **Comparison of Mobile Generations (2G to 5G)**

Generation	Max Speed	Key Features	Impact on Internet Services
2G	9.6 kbps – 14.4 kbps	Voice calls, SMS, basic internet access	Basic email, slow web browsing, text-based social networks
3G	Up to 2 Mbps	Video calls, mobile browsing, multimedia access	Improved email, social media, mobile chat (text & MMS)
4G	100 Mbps – 1 Gbps	High-speed internet, HD video streaming, VoLTE, cloud services	Real-time email, social media, video calls, cloud gaming
5G	Up to 20 Gbps	Ultra-fast data speeds, low latency, IoT, AR/VR, autonomous vehicles	Real-time communication, immersive social media, cloud gaming

PRACTICAL SET – 6

6. Introduction of Git repository and integration with development environment.

- **Git** is one of the most widely used distributed version control systems that help developers work collaboratively on projects. It enables multiple developers to work on different parts of a project simultaneously, tracks every change made to the codebase, and makes collaboration efficient and seamless.
- Git repositories allow developers to store, manage, and track their code changes over time, ensuring that everyone works on the latest version of the project and can revert changes if necessary.
- This study will cover the basics of **Git**, **Git repositories**, and how Git integrates with modern development environments to streamline software development.

➢ Integration of Git with Development Environments

- Git can be integrated with various development environments to facilitate easier version control and code management. Many **Integrated Development Environments (IDEs)** and **text editors** now come with built-in support for Git, which streamlines the process of version control, collaboration, and deployment.

➢ Popular Development Environments with Git Integration:

1. **Visual Studio Code (VSCode)**:
    a) **Built-in Git Support**: VSCode natively supports Git, allowing you to perform Git operations directly from the interface.
    b) **Git Extension**: You can also install extensions like **GitLens** for enhanced Git features, such as visualizing commit history and comparing file changes.
    c) **Source Control View**: The Source Control panel provides a graphical interface to manage your Git repository (staging, committing, pushing, etc.).
2. **JetBrains IntelliJ IDEA**:
    a) **Version Control Tools**: IntelliJ IDEA has Git support built-in, allowing users to clone repositories, commit changes, view branches, and resolve conflicts with a visual interface.
    b) **Git Tool Window**: A dedicated Git tool window allows you to track commits, manage branches, and synchronize with remote repositories.
3. **Eclipse**:
    a) **EGit Plugin**: Eclipse uses the **EGit** plugin to integrate Git into the IDE. It supports cloning repositories, committing changes, and managing branches, all within the Eclipse interface.
    b) **Graphical Interface**: Eclipse provides a graphical interface to visualize the repository structure, history, and branches.

4. **GitHub Desktop**:
    a) **Simplified Git GUI**: GitHub Desktop provides a user-friendly interface to manage Git repositories without needing to use the command line.
    b) **GitHub Integration**: It directly integrates with GitHub, allowing developers to easily clone repositories, commit changes, and push to GitHub without leaving the application.
5. **Sublime Text / Atom**:
    a) **Git Integration via Plugins**: Both Sublime Text and Atom provide Git integration through third-party plugins like **GitSavvy** (for Sublime) and **Git-Plus** (for Atom). These plugins enable Git operations directly from the editor.

➢ Key Git Commands

- git init: Initializes a new Git repository in a directory.
- git clone <repo_url>: Clones a remote repository to the local machine.
- git add <file_name>: Adds a file to the staging area, preparing it for committing.
- git commit -m "message": Commits changes with a message describing what was changed.
- git push: Pushes local changes to the remote repository.
- git pull: Pulls the latest changes from the remote repository to the local machine.
- git branch: Creates and manages branches.
- git merge: Merges changes from one branch into another.

➢ Advantages of Using Git in Development Environment

- **Collaboration**: Git enables teams of developers to work on different features simultaneously, with each working on a separate branch. Changes can be merged later without conflicts.
- **Version Control**: Every change is tracked, and you can easily revert to any previous version of the code. This provides a safety net in case of errors or bugs.
- **Efficient Workflow**: Developers can work offline, and Git will sync the changes when the connection is restored. This flexibility improves the efficiency of development workflows.
- **Conflict Resolution**: Git provides tools to merge changes from different branches and resolve conflicts that may arise when multiple developers modify the same part of the code.

PRACTICAL SET – 7

7. Create/upload files/folders to google drive and share it with your friend.

a. **Create a google document shared with a friend to prepare a report on hardware of a computer system with title page,table of content and proper formatting.**
b. **Create a spreadsheet of findings of surveys on recent technology trends and generate graphs.**
c. **Create a google form for surveys on recent technology trends.**

- Google Drive provides an easy and efficient way to store files in the cloud and share them with others. It allows you to create and manage various types of documents, spreadsheets, and forms. This guide walks through the steps to create and upload files/folders, and how to share them with your friends, as well as how to create a Google document, spreadsheet, and form to collaborate on a project.

➢ Steps for Creating and Uploading Files/Folders to Google Drive and Sharing with a Friend

1. Create a Folder in Google Drive:

1. **Go to Google Drive**:
   - Visit Google Drive and sign in with your Google account.
2. **Create a New Folder**:
   - Click the **"New"** button on the left sidebar.
   - Select **"Folder"**.
   - Name the folder (e.g., "Tech Trends Survey") and click **"Create"**.
3. **Upload Files to the Folder**:
   - Open the folder you just created.
   - Click on the **"New"** button again and select **"File upload"** to upload files from your computer.
   - Alternatively, select **"Folder upload"** to upload an entire folder.
   - Select the files or folders you want to upload and click **"Open"**.
4. **Share the Folder with a Friend**:
   - Right-click on the folder and select **"Share"**.
   - In the "Share with people and groups" box, enter your friend's email address.
   - Choose the permissions: **Viewer**, **Commenter**, or **Editor**.
   - Click **"Send"** to share the folder.

a. Create a Google Document Shared with a Friend to Prepare a Report on the Hardware of a Computer System with Title Page, Table of Contents, and Proper Formatting

- **Steps to Create the Google Document:**

1. **Create a New Google Document**:
   - In Google Drive, click the **"New"** button and select **"Google Docs"**.
   - A blank document will open.
2. **Title Page**:
   - At the top of the document, type the title of your report (e.g., "Hardware of a Computer System").
   - Add your name, date, and other relevant information on the title page.
   - You can center-align the title and adjust the font size and style for emphasis.

   Example:

   Hardware of a Computer System
   Report Prepared by: [Your Name]
   Date: [Date]

3. **Insert Table of Contents**:
   - Go to **Insert > Table of contents**.
   - You can choose a simple format or one with links.
   - This will automatically generate a table of contents based on the headings you use in the document.
4. **Proper Formatting**:
   - Use **Headings** for sections (e.g., **Heading 1** for main sections, **Heading 2** for subsections).
   - For example, structure your report like:
     - **Introduction**
     - **Motherboard**
     - **CPU**
     - **RAM**
     - **Storage Devices**
     - **Graphics Card**
     - **Conclusion**
5. **Share the Document**:
   - Click the **"Share"** button at the top-right of the document.
   - Enter your friend's email address and choose **Editor** permissions so they can add and edit the content.

b. Create a Spreadsheet of Findings of Surveys on Recent Technology Trends and Generate Graphs

➢ **Steps to Create a Google Spreadsheet and Graphs:**

1. **Create a New Google Spreadsheet**:
   - In Google Drive, click the **"New"** button and select **"Google Sheets"**.
   - A blank spreadsheet will open.
2. **Input Data**:
   - In the first row, add the headings for the survey findings. For example:
     - **Technology Trend**, **Percentage Adoption**, **Year**.
   - Below these columns, input the data you gathered from your survey.

- Example:

Technology Trend	Percentage Adoption	Year
Artificial Intelligence	70%	2024
Blockchain	50%	2024
5G Technology	65%	2024
IoT (Internet of Things)	55%	2024

3. **Generate Graphs**:
   - Select the data you want to graph (e.g., the technology trends and percentage adoption).
   - Click on the **"Insert"** menu and choose **"Chart"**.
   - Google Sheets will automatically generate a chart. You can customize the chart type (e.g., bar chart, line chart, pie chart) from the chart editor that appears on the right.
4. **Share the Spreadsheet**:
   - Click the **"Share"** button in the top-right corner.
   - Enter your friend's email address and choose the permissions you want (Viewer, Commenter, or Editor).
   - Click **"Send"**

c. Create a Google Form for Surveys on Recent Technology Trends

➢ **Steps to Create a Google Form:**

1. **Create a New Google Form**:
   - In Google Drive, click the **"New"** button and select **"Google Forms"**.
   - You'll be taken to a new blank form.

2. **Set Up Form Title and Description**:
    - In the top-left, give your form a title (e.g., “Survey on Recent Technology Trends”).
    - Add a description if necessary, explaining the purpose of the survey.
3. **Add Survey Questions**:
    - Click on the **"Untitled Question"** field to add your first question. Choose the question type (e.g., multiple choice, checkboxes, short answer).
    - Example questions could be:
        - **What is the most exciting technology trend for 2024?**
        - **How familiar are you with 5G technology?**
        - **Which technology do you think will have the biggest impact in the next five years?**

    Use **Multiple Choice** or **Dropdown** for options like:

    - Artificial Intelligence
    - Blockchain
    - 5G
    - Internet of Things
4. **Customize the Form**:
    - You can change the theme and color of the form by clicking on the **palette icon** at the top-right corner.
    - Add images or videos if required by clicking on the appropriate icons.
5. **Preview the Form**:
    - Click the **"eye" icon** at the top-right to preview how the form will look to respondents.
6. **Share the Form**:
    - Click the **"Send"** button at the top-right of the form.
    - You can send the form via email or copy the link to share it directly with your friend or survey participants.

PRACTICAL SET – 8

1.1 Create a Presentation regarding recent technology trends.

Step 1: Understand the Assignment

- **Objective**: Make sure you understand the goal of your presentation.

Step 2: Research and Gather Information

- **Identify Recent Technology Trends**: Focus on emerging technologies in the field of Information and Communication Technology (ICT). Some examples:
    - Artificial Intelligence (AI)
    - Cloud Computing
    - 5G Networks
    - Cybersecurity
    - Internet of Things (IoT)
- **Find Reliable Sources**: Use websites, reports, and articles from credible sources such as tech magazines, research papers, and news websites.

Step 3: Create an Outline

- **Title Slide**: Include the presentation title, your name, date, and subject.
- **Introduction Slide**: Provide an overview of technology trends and their significance.
- **Body Slides**: Create individual slides for each technology trend. For each trend, include:
    - Trend Name
    - Key Points (What is it? How does it work?)
    - Applications (How it's being used in real life)
    - Impact (How it changes industries or society)
- **Conclusion Slide**: Summarize the main points and highlight the future of these technologies.

Step 4: Design Your Presentation

- **Choose a Simple Template**: Pick a clean, professional template. Avoid too many flashy colors or fonts.
- **Use Images and Diagrams**: Add visuals to explain the concepts. For example, diagrams for cloud computing or graphs showing data on AI growth.
- **Limit Text**: Use bullet points and keep text short and clear. Don't overload slides with too much information.

Step 5: Add Content to Each Slide

- **Title Slide**: "Recent Technology Trends in ICT"
- **Introduction Slide**: Explain what ICT is and why it's important.
- **Trend Slides**: For each technology trend:
    - Example: **AI and Machine Learning**
        - **Definition**: What is AI/ML?
        - **Applications**: Chatbots, voice assistants, and healthcare diagnostics.
        - **Impact**: Makes processes smarter and more efficient.
- **Conclusion Slide**: Highlight the key trends and their potential impact on the future.

Step 6: Ensure Consistency

- **Font**: Use a simple font like Arial or Calibri. Keep the size readable (e.g., 24 pt for text, 32 pt for titles).

- **Colors**: Stick to a consistent color scheme. Use high-contrast colors for readability (e.g., dark text on light background).
- **Spacing**: Leave enough space between text and images for a clean layout.

Step 7: Review and Edit

- **Proofread**: Check for spelling or grammatical errors.
- **Clarity**: Ensure your slides are clear and easy to understand.
- **Visuals**: Ensure all images and diagrams are clear and relevant to the topic.

Step 8: Practice Your Presentation

- **Rehearse**: Go through your slides and practice presenting. This will help you understand what to say and ensure you don't miss important points.
- **Time Management**: Make sure you can present within the time limit.

Step 9: Save and Export

- **Save Your Work**: Save your PowerPoint file in both .pptx and .pdf formats for easy sharing.
- **Backup**: Save it on a USB drive or cloud storage in case of any technical issues.

Example:



Artificial Intelligence and Machine Learning

AI and ML are transformative technologies reshaping industries, improving decision-making, and solving complex problems.

What is Artificial Intelligence (AI)?

AI is the field of computer science that focuses on creating systems capable of performing tasks that typically require human intelligence. These include:

- Problem-solving
- Understanding language
- Recognizing patterns
- Decision-making

What is Machine Learning (ML)?

ML is a subset of AI that enables computers to learn and improve from data without being explicitly programmed. It focuses on algorithms that:

- Analyze data
- Identify patterns
- Make predictions or decisions

Activate Windows
Go to Settings to activate Windows.

5G and Beyond

5G and Beyond refers to the next-generation of wireless technology, starting with **5G** (the fifth generation) and looking ahead to future developments like **6G**. These technologies promise faster speeds, lower latency, and greater connectivity, transforming how we interact with the world and enabling new technologies.



☐ **Faster Speeds**: 5G can offer download speeds up to **100 times faster** than 4G, enabling quicker data transfer, smoother streaming, and better download experiences.

- **Peak download speed**: 10+ Gbps (compared to 1 Gbps for 4G).

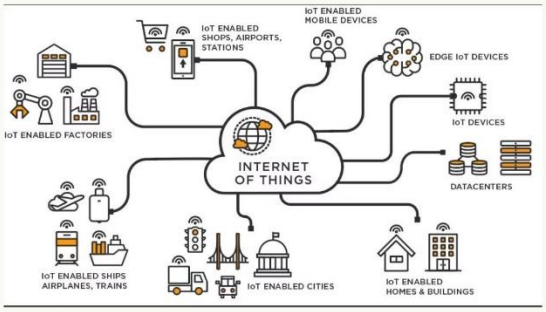
Activate Windows
Go to Settings to activate Windows.

Applications of 5G

- Autonomous Vehicles: Real-time communication for safer and more efficient self-driving cars.
- Smart Cities: 5G enables efficient traffic management, smart lighting, waste management, and environmental monitoring.
- IoT (Internet of Things): A vast number of devices (smart sensors, wearables) connected seamlessly for various uses in industries like agriculture, healthcare, and manufacturing.
- Augmented Reality (AR) and Virtual Reality (VR): 5G's low latency supports immersive experiences, such as VR gaming, remote training, and AR for navigation or shopping.
- Telemedicine: High-speed, low-latency connections make remote surgeries and consultations possible.
- Industrial Automation: In manufacturing and logistics, 5G enables real-time data analysis and automation.

Activate Windows
Go to Settings to activate Windows.

Internet of Things (IoT)

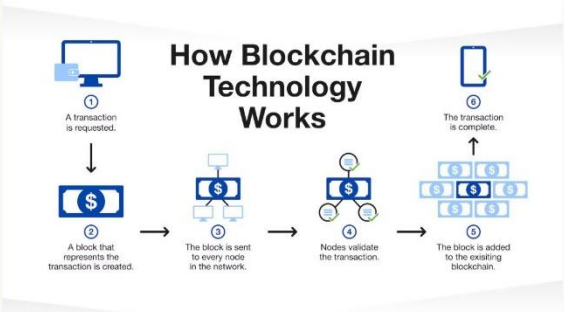


Key Points:

- Increasing adoption in smart homes, healthcare, and industry.
- Integration with AI for predictive analytics.
- Security and privacy challenges in IoT devices.

Activate Windows
Go to Settings to activate Windows.

Blockchain Technology

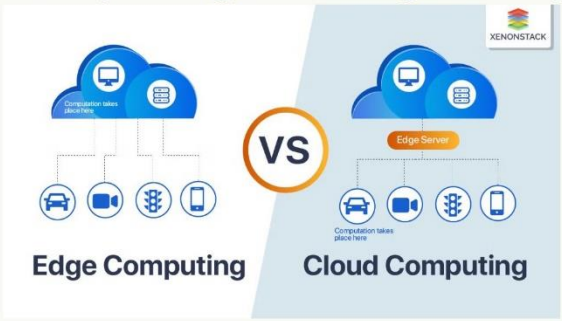


Key Points:

- Beyond cryptocurrency: Applications in supply chain, finance, and healthcare.
- Growth of decentralized finance (DeFi) platforms.
- Emerging trends: NFTs, smart contracts, and green blockchain solutions.

Activate Windows
Go to Settings to activate Windows.

Cloud Computing and Edge Computing



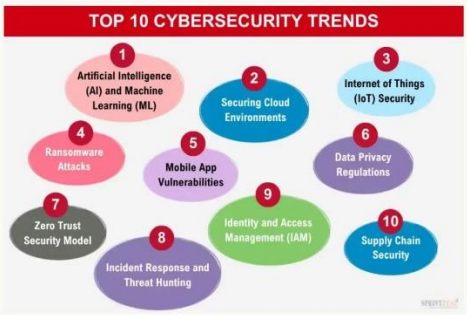
Key Points:

- Expansion of hybrid and multi-cloud strategies.
- Rise of edge computing for low-latency applications.
- Role in supporting AI, IoT, and big data.

Activate Windows
Go to Settings to activate Windows.

Cybersecurity Trends

- Key Points:
  - Growing threat of ransomware and cyberattacks.
  - Adoption of zero-trust architecture.
  - Importance of AI in threat detection and response.
  - Focus on protecting critical infrastructure.



1. **AI and Machine Learning in Cybersecurity**
- **Enhanced Threat Detection**: AI and ML are increasingly used to identify anomalies, detect malware, and predict potential threats before they materialize.
- **Automated Responses**: AI can enable real-time automated incident responses to minimize damage.

Activate Windows
Go to Settings to activate Windows.

2. **Ransomware as a Service (RaaS)**
- **Proliferation of Ransomware**: RaaS platforms make it easy for less-skilled cybercriminals to launch sophisticated ransomware attacks.
- **Double and Triple Extortion**: Attackers threaten to release stolen data publicly or target clients and partners for ransom.

3. **Supply Chain Attacks**
- **Indirect Targeting**: Attackers exploit vulnerabilities in third-party vendors or suppliers to breach larger organizations.
- **Software Supply Chain**: Tools and dependencies used by organizations are increasingly targeted.

Activate Windows
Go to Settings to activate Windows.

Sustainable Technology

Sustainable technology refers to innovations and practices that aim to minimize environmental impact, conserve resources, and promote ecological balance while meeting human needs.

1. **Renewable Energy Technologies**
   - **Solar Power**: Improved solar panels and energy storage systems.
   - **Wind Energy**: Advanced turbines for more efficient power generation.
   - **Hydropower**: Eco-friendly designs that minimize impact on aquatic ecosystems.
2. **Energy Efficiency**
   - **Smart Grids**: Systems that optimize electricity distribution to reduce waste.
   - **Energy-Saving Devices**: Smart appliances and lighting reduce power consumption.
3. **Green Manufacturing**
   - **Circular Economy**: Products designed for reuse, recycling, and reducing waste.
   - **Eco-Friendly Materials**: Use of biodegradable and sustainable materials in production.

Activate Windows
Go to Settings to activate Windows.

Market Impact



Global Transformation

Technology trends reshaping industries worldwide.



Disruption Opportunities

New avenues created by emerging tech trends.



Innovation Acceleration

How trends drive rapid advancements in markets.

Activate Windows
Go to Settings to activate Windows.

The Future of Work

- Key Points:
  - Remote work technologies and collaboration tools.
  - Impact of AI on job automation and reskilling.
  - Evolution of workplace culture and hybrid models.

Activate Windows
Go to Settings to activate Windows.

PRACTICAL SET – 9

1.2 Understand basic documentation with Latex.

Origins of LaTeX: 1980s

- LaTeX was developed by **Leslie Lamport** in 1983 as a macro package for the **TeX typesetting system**, created by **Donald Knuth** in 1978. TeX itself was designed to enable the creation of high-quality printed documents, particularly in academic and scientific settings, with a strong focus on mathematical typesetting.
- LaTeX is a typesetting system commonly used for creating documents that require complex formatting, like academic papers, technical documents, or books. It is especially useful for handling mathematical formulas and symbols.

1. Document Structure

A basic LaTeX document looks like this:

- \documentclass{article} % Specifies the document class (article, report, book, etc.)
- \begin{document} % Start of the document content
- Hello, World! % Text to display
- \end{document} % End of the document content

2. Sections and Subsections

You can structure your document with sections, subsections, and subsubsections.

- \section{Introduction}
  This is the introduction.
- \subsection{Motivation}
  This is the motivation section.
- \subsubsection{Details}
  This is a detailed subsection.

3. Mathematics

LaTeX is well-known for its ability to render complex mathematical equations.

- **Inline math** (within text): Enclosed by single dollar signs.
  This is an inline equation: $E = mc^2$.
- **Display math** (centered and on its own line): Enclosed by double dollar signs or `\[ \]`.
  \[
  E = mc^2
  \]

4. Lists

LaTeX supports both unordered and ordered lists.

- **Unordered List**:
  \begin{itemize}
  \item Item 1
  \item Item 2

\end{itemize}

- **Ordered List**:

\begin{enumerate}
\item First item
\item Second item
\end{enumerate}

5. Tables

Tables can be created using the `tabular` environment.

\begin{tabular}{|c|c|c|}
\hline
Header 1 & Header 2 & Header 3 \\
\hline
Cell 1 & Cell 2 & Cell 3 \\
\hline
\end{tabular}

6. **Figures**

You can include images using the `graphicx` package.

- \usepackage{graphicx} % Add this line in the preamble (before \begin{document})
- \begin{figure}[h]
- \centering
- \includegraphics[width=0.5\textwidth]{image.jpg} % Replace with your image file path
- \caption{Example figure.}
- \end{figure}

7. **Comments**

Any text preceded by a `%` is considered a comment and will not be displayed in the final document.

- % This is a comment and will not appear in the document

8.Compiling

To compile a LaTeX document:

1. Write your LaTeX code in a `.tex` file.
2. Use a LaTeX editor (like Overleaf or TeXShop) to compile it into a PDF.

PRACTICAL SET – 10

1.3 Study different AI and ICT tools.

AI (Artificial Intelligence) and ICT (Information and Communication Technology) tools are vast and continually evolving.

1. AI Tools

AI tools are designed to automate tasks, analyze data, and mimic human cognitive functions. Some common categories include:

Machine Learning Tools

- **TensorFlow**: An open-source machine learning framework by Google that helps in building deep learning models.
- **PyTorch**: A deep learning framework that provides flexibility and is widely used in research.

Natural Language Processing (NLP) Tools

- **GPT (Generative Pretrained Transformer)**: Language models like GPT-3, GPT-4, and their successors are used for chatbots, text generation, summarization, and translation.
- **spaCy**: A library for advanced NLP in Python, used for text processing, named entity recognition, and text classification.

Computer Vision Tools

- **OpenCV**: A popular computer vision library used for real-time image processing and analysis.
- **DeepFace**: A lightweight face recognition library in Python that provides easy-to-use face recognition models.

AI for Automation

- **UiPath**: A robotic process automation (RPA) tool used to automate repetitive business processes.
- **Blue Prism**: Another RPA tool used for automating business processes with AI capabilities.

2. ICT Tools

ICT tools encompass a broad range of technologies that help in data communication, computing, and internet services.

Productivity and Collaboration Tools

- **Microsoft Office 365**: A suite of productivity tools like Word, Excel, PowerPoint, and Teams, widely used for document management and collaboration.
- **Google Workspace**: A cloud-based productivity suite including Google Docs, Sheets, Drive, and Meet for collaboration and document management.
- **Slack**: A team communication platform that integrates with numerous apps to streamline team collaboration.

Cloud Computing Tools

- **Amazon Web Services (AWS)**: A comprehensive cloud computing platform that provides services for computing, storage, and databases.

- **Microsoft Azure**: A cloud computing service offering infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS).
- **Google Cloud**: Provides cloud storage, computing services, and machine learning tools.
- **IBM Cloud**: A cloud platform that combines infrastructure as a service, platform as a service, and software as a service.

Data Analytics and Business Intelligence Tools

- **Tableau**: A powerful tool for data visualization and business intelligence that allows users to create interactive dashboards.
- **Power BI**: A business analytics tool by Microsoft that enables the visualization of data from multiple sources.
- **Google Analytics**: A web analytics service that tracks and reports website traffic.
- **Hadoop**: A framework used for processing large datasets in a distributed computing environment.
- **Apache Spark**: A fast and general-purpose cluster computing system for big data processing.

Cybersecurity Tools

- **Wireshark**: A network protocol analyzer used to capture and analyze packets in real-time.
- **Kali Linux**: A Debian-based distribution with numerous tools for penetration testing and ethical hacking.

Networking Tools

- **Cisco Packet Tracer**: A network simulation tool used to create network topologies and simulate network traffic.
- **Wi-Fi Analyzer**: Tools for testing and troubleshooting wireless networks, ensuring optimal connectivity.

Website Development and Design Tools

- **WordPress**: A content management system (CMS) for building websites and blogs.
- **Wix**: A website builder with drag-and-drop functionality, ideal for users without coding skills.
- **GitHub**: A platform for version control and collaboration, allowing developers to store and manage code repositories.
- **Figma**: A cloud-based design tool for UI/UX design, used for wireframing and prototyping